Installation of MySQL

The first step of this assignment is to install MySQL Server Community Edition.

Here are a couple of videos that cover this task. The installation comes with the database management tool (DBMS) MySQL Workbench.

Windows: https://www.youtube.com/watch?v=GIRcpjg-3Eg
Mac: https://www.youtube.com/watch?v=n1zT1OZcgnw

I expect you to be able to use a search engine, YouTube video or some other way to be able to figure things out. If these videos above do not work for you, then exercise your search capabilities and find one that does work. Also, please use Discord for questions and help. Many times classmates can assist with a problem they may have previously resolved. I also monitor these channels closely and will respond as soon as possible. Posting screen shots, error messages, steps you followed is incredibly helpful. We need as much information as possible.

We will be using the MySQL shell (command line interface) and also working with the Workbench graphical tool. It is important to be familiar with both the command line and the graphical interface when dealing with servers. You do not always have the luxury being able to call up a nice full featured GUI environment for server maintenance. Sometimes you will be logging into a remote server over SSH and only have a terminal window to execute commands. This is an important IT skill.

Logging into MySQL from the terminal: Mac OSX and Windows

If you are a Linux user, the Mac material should be similar enough to get you going. If not, use a search engine or jump on Discord for help

This assignment assumes you are familiar with basic command line navigation and possess a general understanding of your file system. If you feel you need to do a little review check these videos out.

- Windows Command Line: **https://www.youtube.com/watch?v=MBBWVgE0ewk**
- Mac OSX: **https://www.youtube.com/watch?v=aKRYQsKR46I**
- If you have any questions hop on over to Discord. Don't be shy

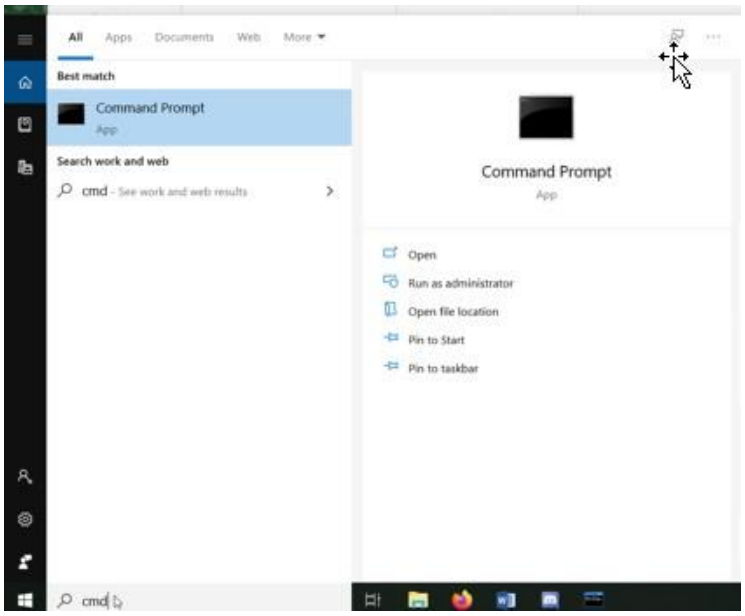
I will be asking you to take screen shots of your progress for submission. The screen shots should be added to a document that can then be submitted to blackboard. You can use whatever word processor you prefer but please export the document as a PDF.

The regular old screenshot systems that come with your OS aren't that great. I much prefer

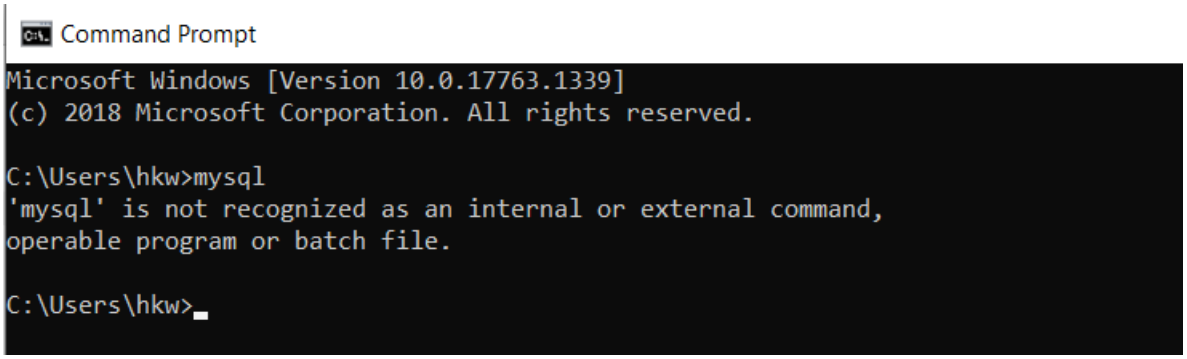
- **Greenshot on Windows: https://getgreenshot.org/**
- **Flameshot on Mac OSX: https://flameshot.js.org/#/**

Logging into MySQL from the Windows Command Line (keep scrolling for Mac OSX instructions)

1. The first task here is to determine if Windows command line "knows" where MySQL was installed. Open a command prompt window by typing **cmd** in the Windows Search Bar. I like to pin this to the Task Bar for easy retrieval.

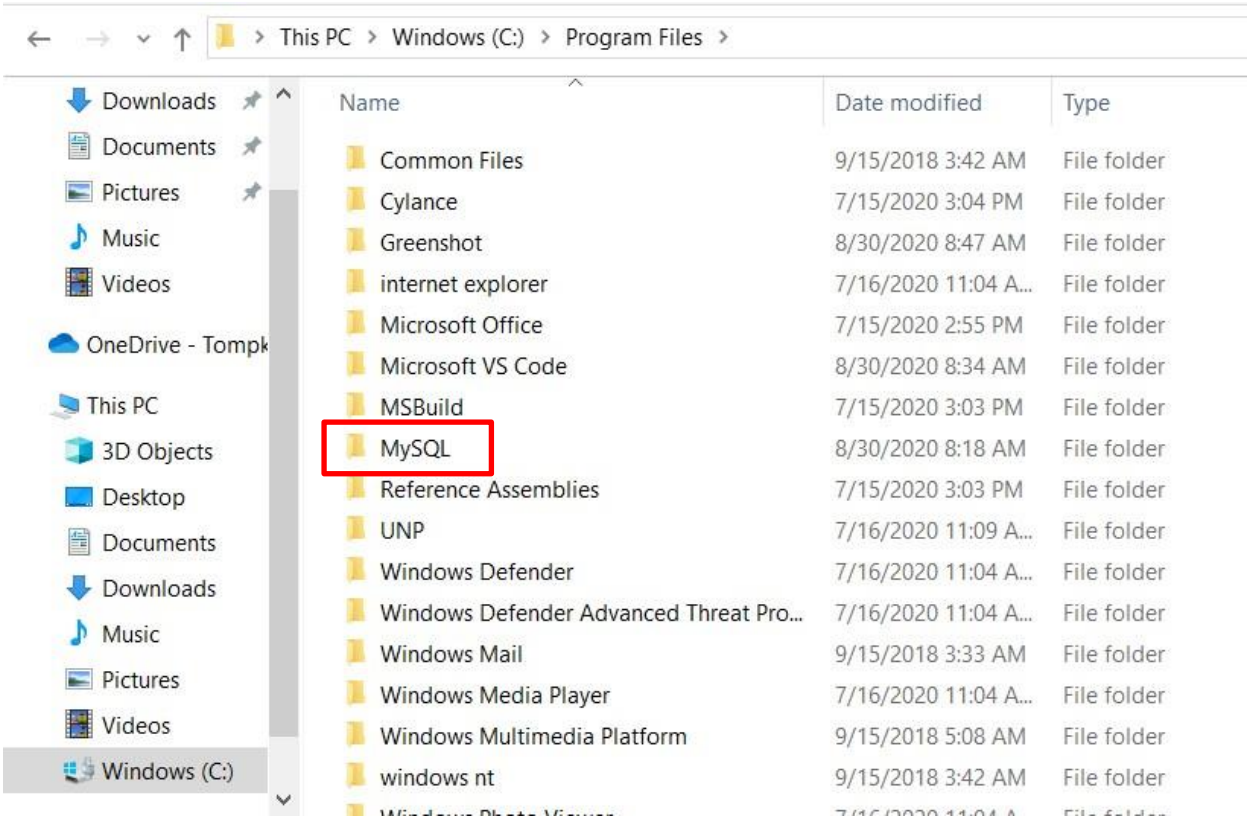


2. At the prompt type the command **mysql**. The results of this command will answer our question from number 1. As expected, I get the **Command Not Found Error** as shown here.

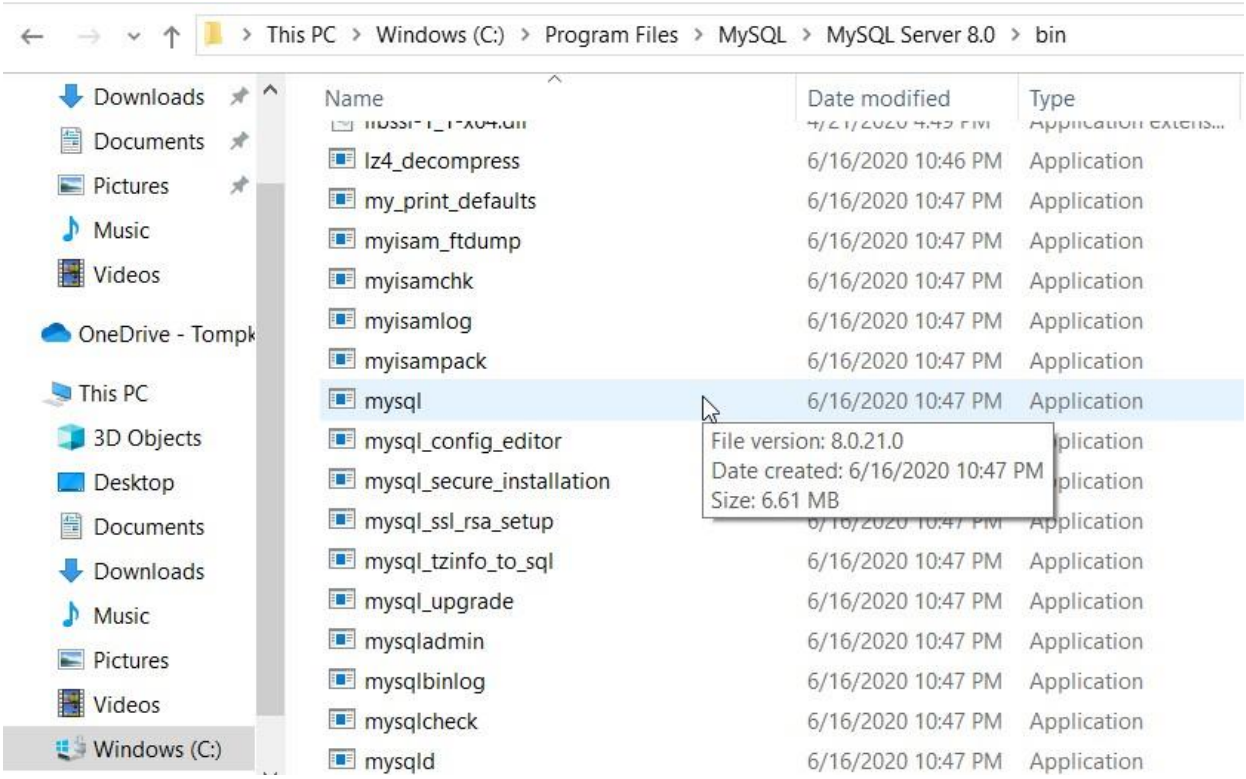


   This is not a deal breaker. We just need to inform command prompt of the MySQL installation ***path***

3. Where was MySQL installed? The default installation path on Windows is usually: **C:\Program Files**

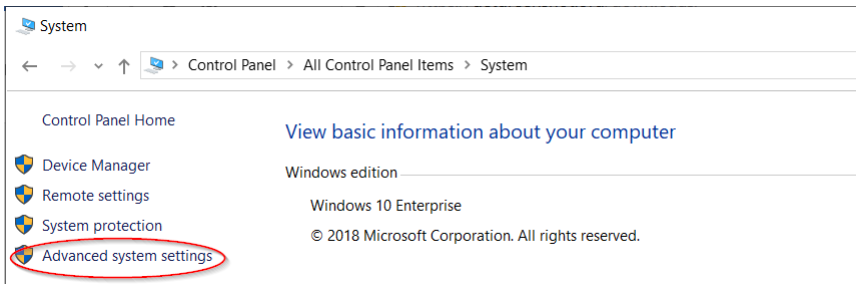


4. Navigate into the MySQL directory. You will find a sub-directory called**: MySQL Server 8.0**. (your numbering may be different. That is fine.) Navigate inside this directory and you will find a sub-directory called: **bin**. "Bin" is short for ***binary*** and is the directory where all MySQL executable programs are stored. I have highlighted the ***mysql*** command we attempted to execute from the command line.

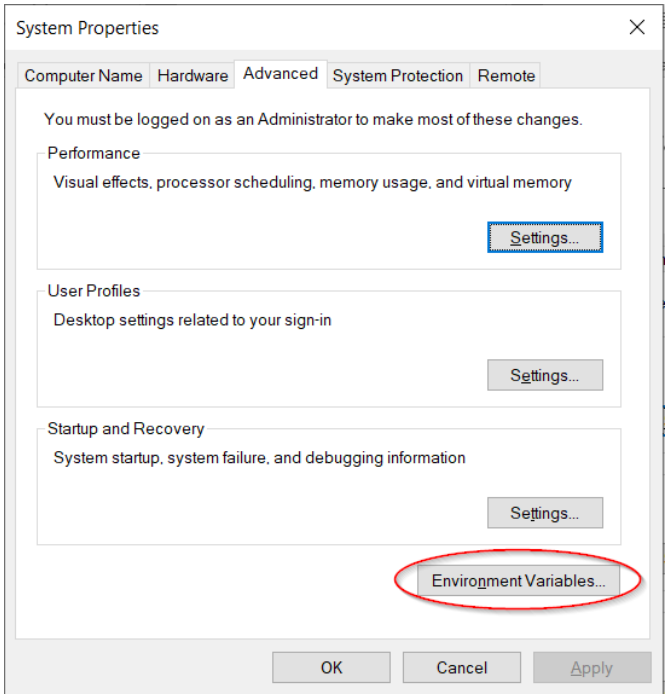


5. Now we need to take that ***installation path*** discovered above and put it somewhere that Windows will ***search*** through whenever a command is executed from the command line. This location is called an ***environment variable***, and oddly enough the variable is called **Path.** This variable contains paths to a variety of directories on your system and Windows (and Mac and Linux) will scan this location every time you enter a command at the prompt, trying to locate the associated program to execute. We need to copy that installation path into the Path environment variable. Then we will be able to execute that command from the prompt. This is not specific to MySQL and is a command system configuration.

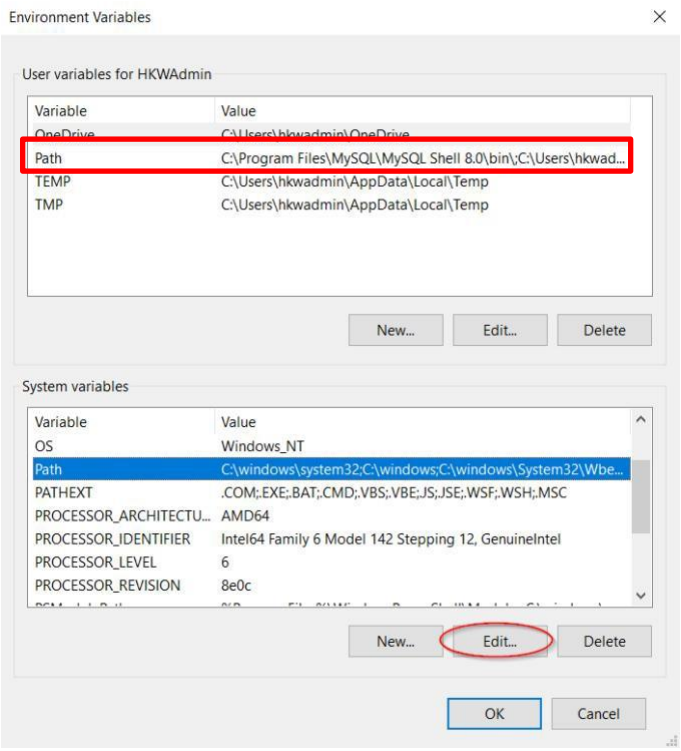
6. Open **Control Panel** and click on **System**. That will bring up this window. Select **Advanced System Settings**. You may have to enter your administrator password.



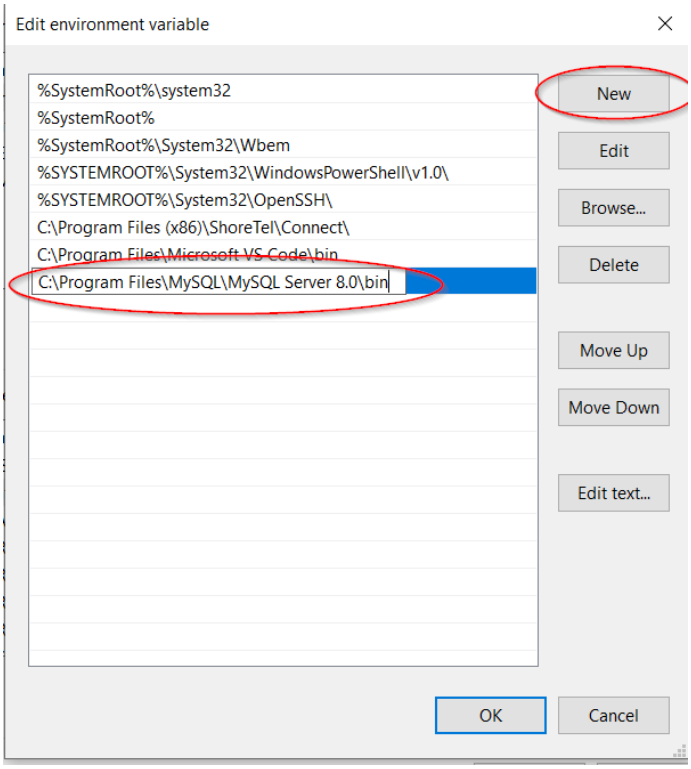
7. You should now see this window. Select **Environment Variables**



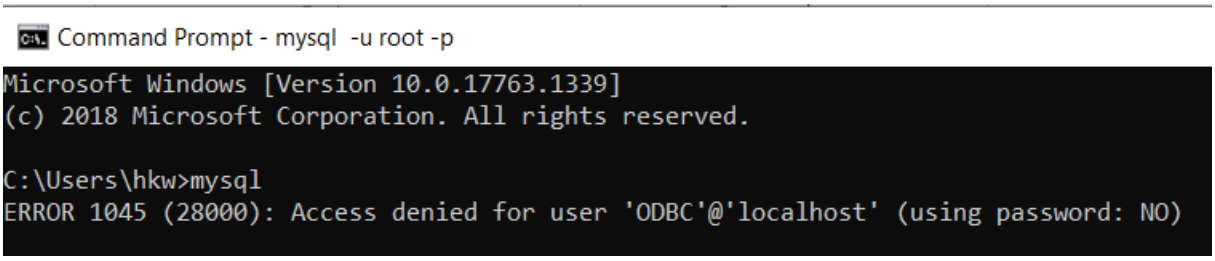
8. When the Environment Variables window opens go to the bottom pane labeled **System Variables** and locate the **Path** variable and select **Edit.**



9. The following window will appear. Click the **New** button and copy the MySQL installation path from above into the highlighted area. Make certain that you incude the bin folder in the path. Your window may look different from mine. Don't worry. Click **OK** out of all of the system windows.



10. Restart the command prompt so its **memory** can be refreshed with the new addition to its environment. Type the **mysql** command again. This time you should get some different results.



This is still an error, but it is a ***different error***. A connection to the MySQL server is being attempted, but access is denied because we did not identify ourselves. This does let us know that our configuration worked.

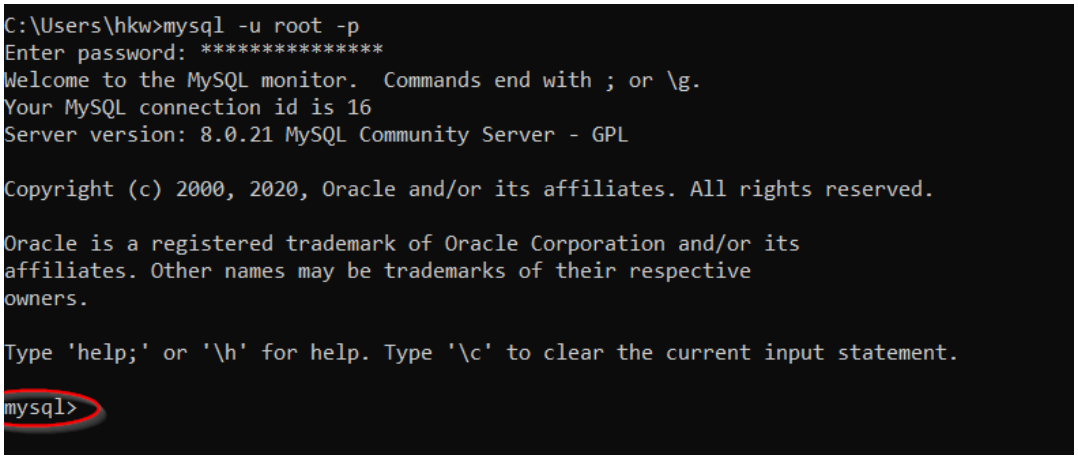
11. Let's now login correctly. We need to log in as the ***root user***. This is accomplished with the following command

    mysql –u root –p

    The "u" switch allows us to login to MySQL as a particular user. The "p" switch tells MySQL that we'd like to enter our password.

    Type this at the command prompt and enter your ***MySQL root password*** when prompted. You'll notice that the password is being obscured. MySQL actually opened an encrypted channel for you to enter your password. This prevents shoulder surfing and isquite secure. Read the output here. You'll see that you are being welcomed into the MySQL Monitor. Notice how the prompt has changed. While the monitor is running youcan only execute MySQL commands. Attempting to enter Windows command prompt commands will result in an error. This is an important concept to understand.

    Screenshot this window for your submission. Be sure to include the C:\ prompt so I can tell that it is you. Add the screen shot to the submission document



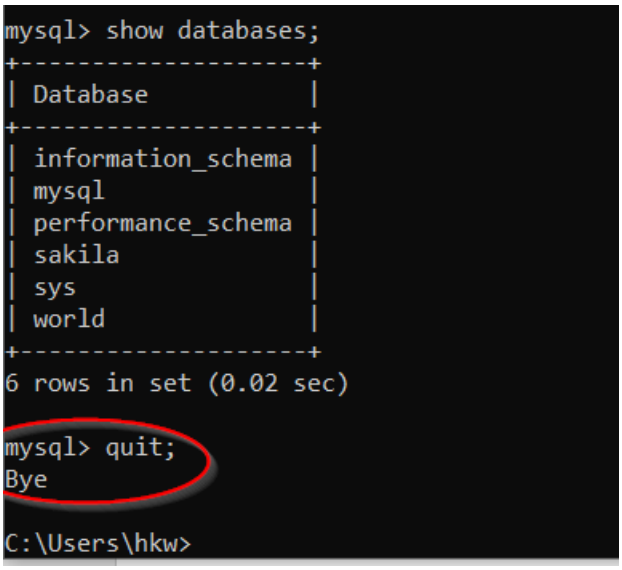
12. At the **mysql>** prompt enter the command **show databases;** you should see a collection of databases printed to the window.

Also take a screen shot of these results and add it to your submission document

```
mysql> show databases;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| mysql              |
| performance_schema |
| sakila             |
| sys                |
| world              |
+--------------------+
6 rows in set (0.02 sec)

mysql>
```

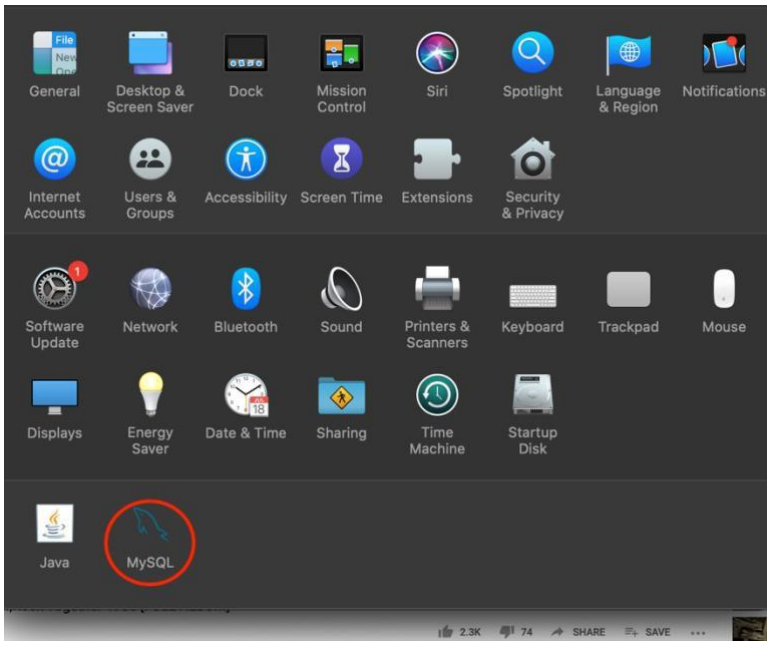
13. Now enter the command **quit;** this will exit the MySQL Monitor and take you back to the Windows command prompt. Notice how the prompt changes. Now you are back to entering **Windows commands.** It is important that you keep this straight.



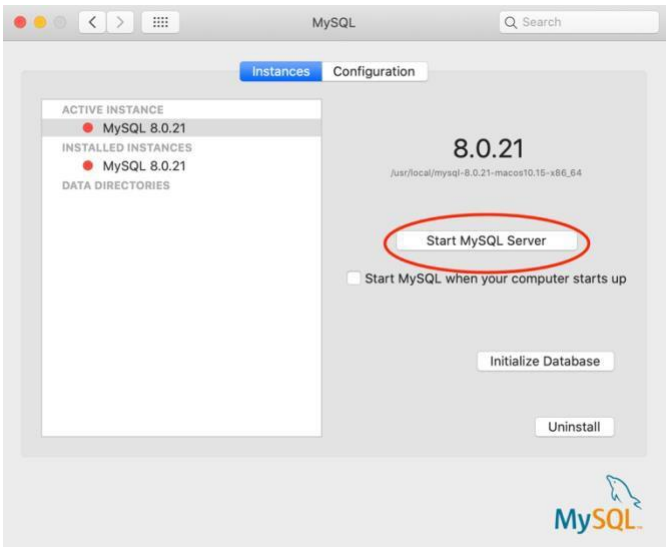
Navigate to the end of this document to locate the final submission instructions.

Logging into MySQL from the Mac OSX Terminal

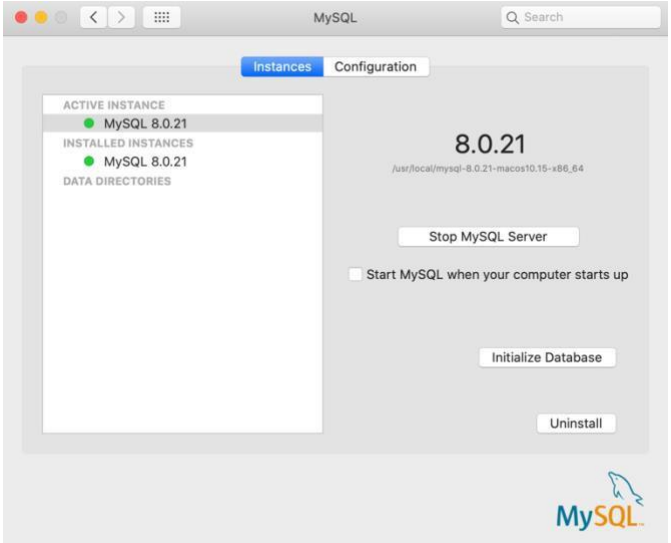
1. Before you can connect to the MySQL server you must make sure that the server has been started. The MySQL installation routine creates a shortcut in ***System Preferences***. Click on the MySQL icon.



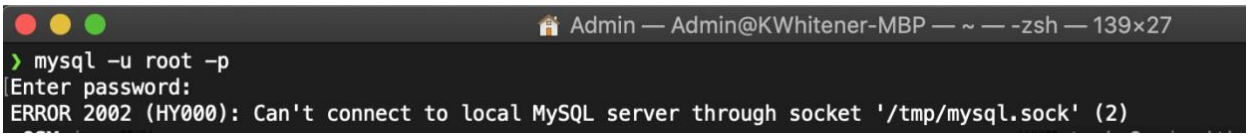
2. You will now see a MySQL control pane. Click the **Start MySQL Server** button. You can probably guess what this is going to do. You will have to do this each time you want to work with the server, unless you click the **Start MySQL when your computer starts up** check box. This is up to you, but my opinion is that you don't need to have unnecessary software running on your development machine. Obviously if this was a mission critical server that we were configuring we would want the server to constantly be running.



You will need to enter your computers password. Notice how the window changes once you have started the server. The green bullets next to MySQL 8.0.21 means it is running and if you press the button again you will be stopping the server. Use this control panel to start and stop the service

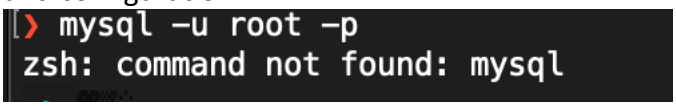


If you ever get the following error message when attempting to login to the server, it is probably due to it not being started.



3. Now that the server is started, let's try to log into to MySQL. Open a terminal and enter the following command at the prompt:
    a. **mysql -u root -p**
    b. This command is requesting to start mysql as the root user and using a password
    c. If you see a window like what is shown below, you were successful. Take a screenshot of this window to show me you were able to accomplish this.

4. If you get the following error message, then you will have to do some further navigation and configuration.



Luckily this is no big deal and is a super helpful thing to know how to do. You are seeing this error because the operating does not know where the mysql executable is located within the file system, so when you invoke the command from the command line the OS doesn't know where to find it. We need to educate OSX on where the program is located

a. **Step One:** Locate the mysql executable. If you didn't do anything crazy during the installation, mysql was installed to **/usr/local/mysql** let's see.

```
 OSX | ~
❯ cd /usr/local
 OSX | ⌀ /usr/local
❯ ls
Caskroom                        etc                             opt
Cellar                          include                         sbin
Frameworks                      lib                             share
Homebrew                        mysql                           var
bin                             mysql-8.0.21-macos10.15-x86_64
```

b. **Step Two:** Looks like we found it. Navigate into the mysql directory at **/usr/local/mysql**. If you show a listing in this location, you will see the all-important **bin** (binary) directory. **Bin** directories are traditionally where executable programs are stored. If you navigate into this directory and do a directory listing, you will see all of the associated ***mysql executables*** and you should the mysql executable itself.

```
❯ cd bin
 OSX | ⌀ /usr/local/mysql/bin                                    with Admin@KWhitener-MBP
❯ ls
ibd2sdi                 mysql                   mysqlbinlog             mysqlpump
innochecksum            mysql_config            mysqlcheck              mysqlshow
lz4_decompress          mysql_config_editor     mysqld                  mysqlslap
my_print_defaults       mysql_secure_installation mysqld_multi          perror
myisam_ftdump           mysql_ssl_rsa_setup     mysqld_safe             zlib_decompress
myisamchk               mysql_tzinfo_to_sql     mysqldump
myisamlog               mysql_upgrade           mysqldumpslow
myisampack              mysqladmin              mysqlimport
```

c. **Step Three:** Because we navigated directly to where the program is installed, we can now login to mysql with the following command. The **./** at the beginning of this command is important

   i. **./mysql -u root -p**

Type this at the command prompt and enter your ***MySQL root password*** when prompted. You'll notice that the password is being obscured. MySQL actually opened an encrypted channel for you to enter your password. This prevents shoulder surfing and is quite secure. Read the output here. You'll see that you are being welcomed into the ***MySQL Monitor***. Notice how the prompt has changed. While the monitor is running you can only execute MySQL commands. Attempting to enter Mac Terminal commands will result in an error. This is an important concept to understand.

```
OSX | ⌀ /usr/local/mysql/bin ──────────────────────── took 7s | with Admin@KWhitener-MBP |
❯ ./mysql -u root -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 8
Server version: 8.0.21 MySQL Community Server - GPL

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql>
```

Screenshot this window for your submission. Add the screen shot to your submission document.

5. At the **mysql>** prompt enter the command **show databases;** you should see a collection of databases printed to the window. Don't worry if you see a different list of databases as that is not the point. You can exit mysql by typing the command **quit;** This will take you back to your Mac Terminal and you can resume entering Mac terminal commands.

```
mysql> show databases;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| mysql              |
| performance_schema |
| sakila             |
| sys                |
| world              |
+--------------------+
6 rows in set (0.02 sec)

mysql>
```

Also take a screen shot of these results and add it to your submission document

6. If you'd like to configure your machine so that you will not have to navigate to the **bin** directory every time you want to login to mysql research the following.

    - The **PATH** environment variable
    - How to add a search location to your **PATH**

- **https://apple.stackexchange.com/questions/176786/how-to-add-mysql-to-path-variable-to-resolve-mysql-command-not-found**

SUBMISSION: Export your submission document to PDF format and attach it to the assignment in Blackboard. You should review the instructions to ensure you have met all of the requirements.